

Answers

Exam in Introduction to economics for sport management

Instructions (read carefully)

- The final exam lasts 3 hours.
- You can use calculators.
- Justify all your answers using economic concepts, theories and reasoning.

Good Luck!

Question 1 (15 points)

Suppose there is a country that can produce two goods: X and Y. The country has 400 machines and 200 workers. Each machine can produce 1 unit of X or 2 units of Y. Each worker can produce 4 units of X or 4 units of Y.

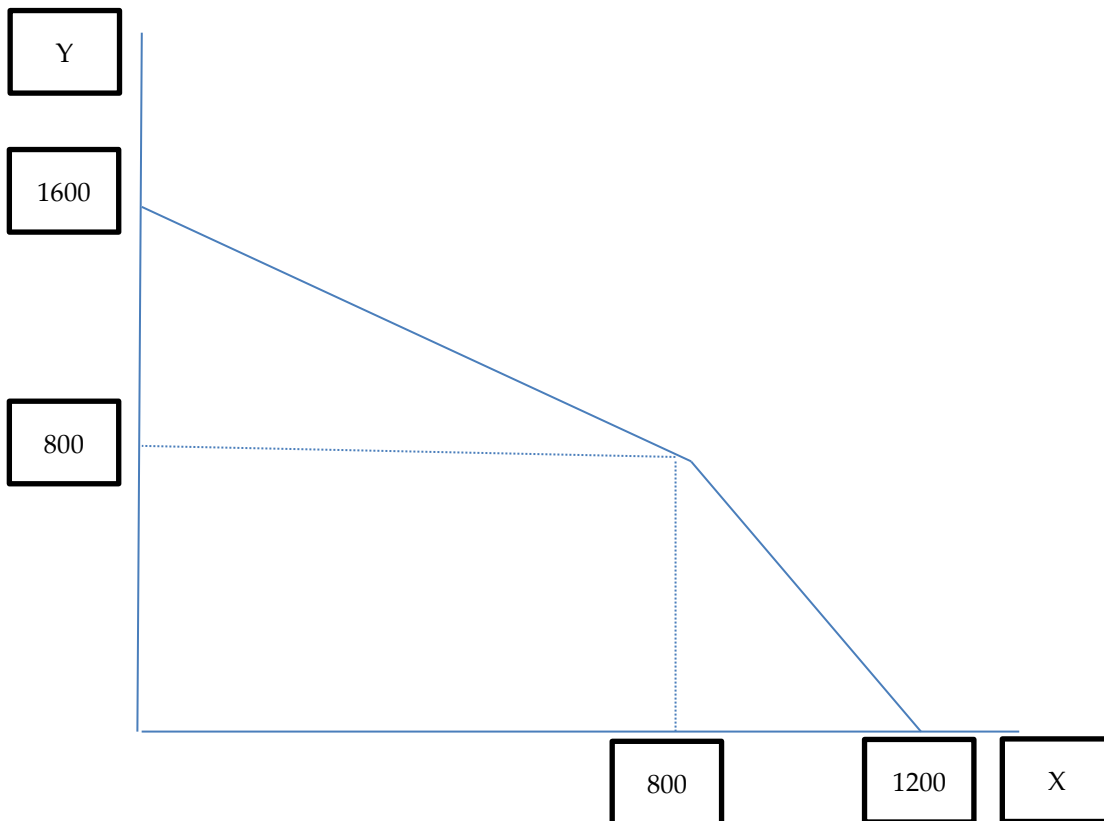
- Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(6 points)**
- Suppose there is a world trade, such that 1 unit of X is worth 3 units of Y. How many units of X and Y should this country produce and why? **(4 points)**
- Suppose that this country needs to consume 1000 units of X. In that case how many units of Y will it consume? **(5 points)**

Answers:

	machines	workers
Production X per one unit	1	4
Production Y per one unit	2	4
MC in producing X in terms of Y	2	1
MC in producing Y in terms of X	0.5	1
Max X	$400 \cdot 1 = 400$	$200 \cdot 4 = 800$
Max Y	$400 \cdot 2 = 800$	$200 \cdot 4 = 800$

- See answer a below.

- b. This country has a comparative advantage in producing X, because in the worst case, its cost of producing X is 2Y. However, they can sell it to the world for the price of $1X=3Y$. Therefore, this country will only produce 1200X and zero Y.
- c. If a country needs to consume 1000 units of X, then it will still produce 1200 units of X. The country will exchange the remaining 200 units of X for $200 \cdot 3 = 600$ units of Y.



Question 2 (20 points)

There is a market with 8 workers who should be allocated between one field A and two fields B with the following production function:

Field A

L	TP
0	0
1	7
2	13
3	16
4	18

Field B

L	TP
0	0
1	3
2	5
3	6

- How many workers would work on each field (on one field A and on each of the two fields B) if it were known that the price of one produced unit on field A is 1USD and the price of one produced unit on field B is 5USD? **(6 points)**
- What would be the salary of each worker? **(6 points)**
- What is the profit of each of the fields' owners? **(4 points)**
- What would be the salary if we add another field A (so there will be two fields A, two fields B and 8 workers)? **(4 points)**

Answer:

Field A

L	TP	mp	Vmp (p*mp)
0	0		
1	7	7	7
2	13	6	6
3	16	3	3
4	18	2	2

Field B

L	TP	mp	Vmp (p*mp)
0	0		
1	3	3	15
2	5	2	10
3	6	1	5

- There will be 3 workers in each of the fields B and 2 workers on field A.
- The value of the marginal product of the last worker is 5 and this will be the wage as well (VMP=W).
- Profit of each owner= $TP \cdot p - W \cdot L$
 Profit of the owner of field A= $13 \cdot 1 - 5 \cdot 2 = 3$
 Profit of each of the owners of field B= $6 \cdot 5 - 5 \cdot 3 = 15$
- If we add another field A, the last two workers that were previously allocated to field B, will now work as worker number 2 one of the fields A, with Vmp=W=6

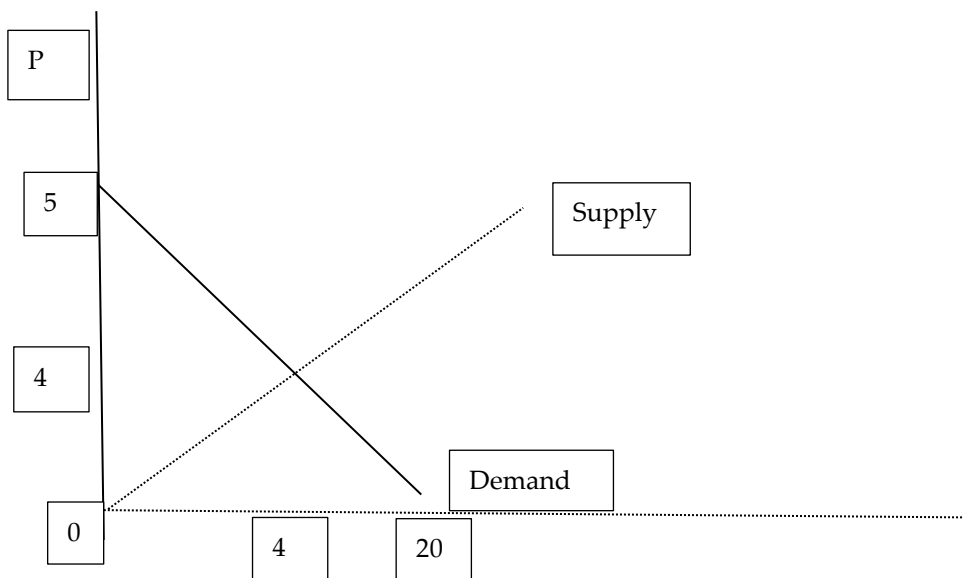
Question 3 (20 points)

Suppose the following demand function of one consumer: $q^D=10-2p$. In addition, suppose the following supply function: $Q^S=p$ where q^D is the quantity demanded of a good of a single consumer, Q^S is the supplied quantity of a good in the market and p is the price of a good in USD. Suppose there are two consumers.

- What is the aggregate demand function? **(4 points)**
- What would be the price and the quantity in equilibrium? **(4 points)**
- Show on the graph (where appropriate) the cases where $p=0$, $Q=0$, and price and quantity in equilibrium. **(8 points)**
- What is the total social welfare? **(4 points)**

Answer:

- The aggregate demand function is $Q^D = 2 * q^D = 2 * (10 - 2p) = 20 - 4p$
- In equilibrium the supply is equal to demand ($Q^S = Q^D$) therefore, $20-4p=p$.
Solving this equation we get $p=4$ and $Q=4$.
- See the answer below.
- The total social welfare is the sum of consumer and producer surplus.
The consumer surplus is the area between the demand curve and the price (triangle area) $(5-4)*(4-0)/2=2$
The producer surplus is the area between the supply curve and the price (triangle area) $(4-0)*(4-0)/2=8$
The total social welfare is $8+2=10$



Question 4 (15 points)

A firm produces when its $mc=20$. It is known that AVC at the production point is 17.

- What is the price according to which a firm sells its production? **(5 points)**
- What is a firm's profit in the short term if it produces 5 units? **(5 points)**

c. What is the maximal ATC, so that a firm will produce in the long term? (5 points)

Answers:

- a. A firm produces at a point where $p=mc$. Therefore, the price is 20.
- b. $20 \times 5 - 17 \times 5 = 15$
- c. The ATC should be lower than the price (20)

Question 5 (10 points)

You are expected to invest a one-time sum of 1500 NOK at the end of the second year from now, another one-time sum of 2000 NOK at the end of the fifth year from now and receive it at the end of the nine years from now. What is the Future Value of this investment (after 9 years from now) if the annual interest rate is 8% in the first year, 5% in the second year, 7% in the third year and 5% in the fourth and the fifth years, and 10% per year after that?

Answer:

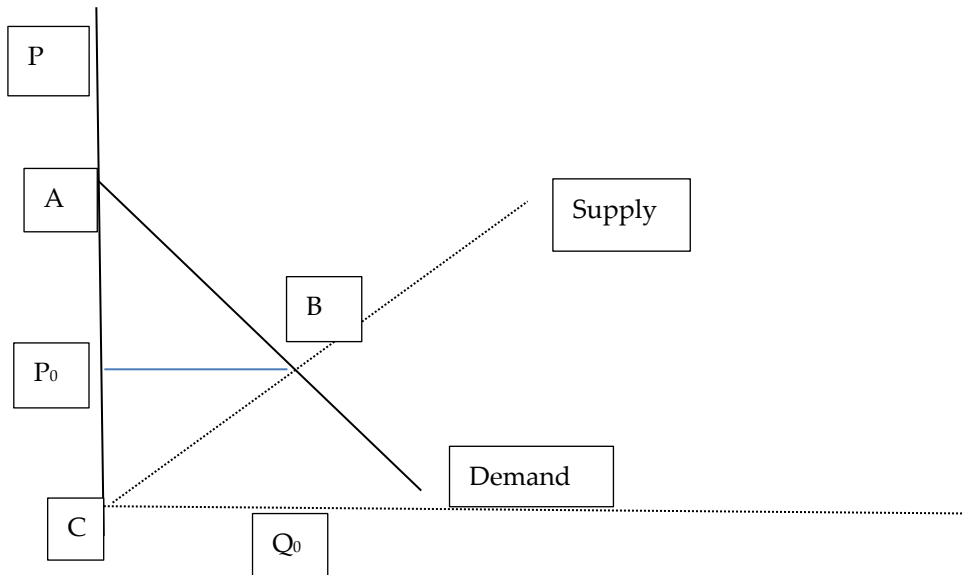
$$FV = 1500 \cdot (1.08)^2 \cdot (1.05)^2 \cdot (1.1)^4 + 2000 \cdot (1.1)^4 = 5519$$

Question 6 (10 points)

Show on graph the loss of social welfare due to subsidy.

Answer:

Before subsidy

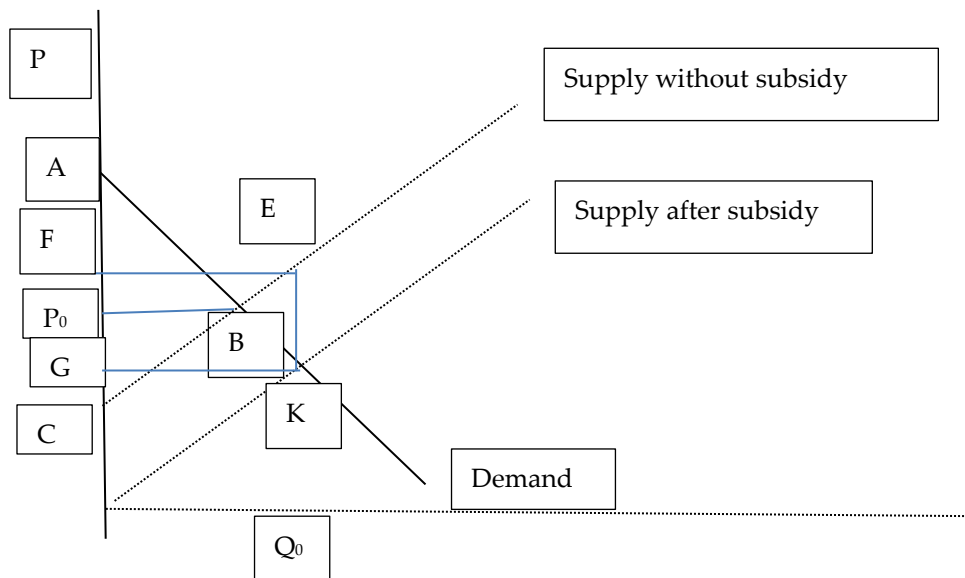


Consumer surplus: ABP_0

Supplier surplus: P_0BC

Total Social welfare: ABC

After Subsidy



Consumer surplus: AKG

Supplier surplus: CEF

Governmental expenditure from subsidy: FEKG

Total Social welfare: ABC- EKB

Loss of welfare: EKB

Question 7 (10 points)

Suppose you are a financial manager of a club that participates in the UEFA Europa League. Suppose that all the home games of your team are scheduled at 21:00 CET.

- Would you anticipate higher or lower attendance compared to if the game would start at 18:45? **(5 points)**
- Would you recommend the owner of the club to increase or decrease the tickets price for these Europa League games and why? **(5 points)**

Answer:

- The attendance at 21:00 is lower compared to 18:45.**
- Given that the attendance is lower (fewer fans), it is likely that the home advantage is lower, and thus a lower probability to gain points, which are worth even more (points, qualification). Thus, it is worth to decrease the price for tickets in order to attract more fans to increase the home advantage.**